

Do urbanization tasks need both views?

4-model sweep — full vs blind

4-model sweep · full vs blind · 9 urbanization tasks

task	Gemma-4-12B		Qwen2.5-VL-7B		Qwen3.5-4B		Qwen3.5-9B	
	full	Δ bl	full	Δ bl	full	Δ bl	full	Δ bl
Land use	47.3	+29.5	55.6	+1.2	55.6	+17.4	53.2	+14.5
Building height	61.6	+29.0	57.9	+13.2	61.6	+15.3	56.8	+11.0
Urban density	37.3	+14.7	41.8	+16.9	36.3	-11.4	39.0	-10.9
Junction type	66.5	+59.0	57.5	+10.5	50.0	+3.0	55.7	+14.1
Green space	58.6	-17.8	81.8	+3.0	41.4	-12.3	36.9	-24.7
Amenity richness	35.6	+11.4	31.8	+17.5	35.4	+16.4	36.3	+15.2
Road type	65.3	+50.3	50.6	+25.2	57.0	+25.9	53.0	+13.1
Road surface	78.2	+51.8	90.8	+8.0	91.4	+3.9	85.5	+32.7
Transit density	28.0	+1.2	28.0	+1.4	34.2	+7.1	31.4	+5.0

full = both views · Δ bl = full — blind (vision gain) · thinking off · n=190–421/task

Single-model — all four views

Qwen3.5-9B · full / sat / sv / blind · 9 urbanization tasks

task	full	sat	sv	blind	full—best1
Land use	53.0	50.1	52.7	43.5	+0.3
Building height	61.1	47.9	64.7	45.8	-3.6
Urban density	38.5	39.2	44.9	40.4	-6.4
Junction type	62.6	48.8	62.6	48.5	+0.0
Green space	36.5	35.5	37.9	64.5	-1.4
Amenity richness	36.8	36.1	33.5	19.5	+0.7
Road type	47.5	40.1	57.5	34.9	-10.0
Road surface	82.2	78.9	87.1	67.7	-4.9
Transit density	32.1	30.9	30.4	27.3	+1.2
overall	48.6	44.4	50.9	40.9	-2.3

sat = satellite only · sv = street-view only · full—best1 = full — max(sat, sv) · thinking off

Cross-view tasks — all four views

Cross-view tasks · full / sat / sv / blind (1/2)

task	model	full	sat	sv	blind	full—best1
Mismatch binary (easy)	Gemma-4-12B	79.8	43.9	52.0	46.6	+27.8
	Qwen2.5-VL-7B	59.1	53.2	53.7	47.5	+5.4
	Qwen3.5-4B	74.3	51.1	51.1	50.6	+23.2
	Qwen3.5-9B	87.2	47.5	54.2	45.1	+33.0
Mismatch binary (hard)	Gemma-4-12B	64.6	54.6	47.5	55.8	+10.0
	Qwen2.5-VL-7B	45.6	44.7	44.2	52.3	+0.9
	Qwen3.5-4B	58.0	47.0	46.3	46.6	+11.0
	Qwen3.5-9B	72.9	55.1	44.9	54.4	+17.8
Mismatch MCQ (easy)	Gemma-4-12B	43.2	21.9	27.3	0.7	+15.9
	Qwen2.5-VL-7B	39.7	24.9	27.3	24.9	+12.4
	Qwen3.5-4B	45.8	25.9	25.2	26.1	+19.9
	Qwen3.5-9B	51.3	23.5	23.3	27.1	+27.8

full—best1 = full — max(sat, sv) · camera_direction has no sv_only (sv = -)

Cross-view tasks · full / sat / sv / blind (2/2)

task	model	full	sat	sv	blind	full—best1
Mismatch MCQ (hard)	Gemma-4-12B	34.7	25.2	24.0	0.7	+9.5
	Qwen2.5-VL-7B	33.7	25.9	24.0	26.4	+7.8
	Qwen3.5-4B	38.5	24.2	25.4	24.9	+13.1
	Qwen3.5-9B	39.4	21.1	21.6	24.2	+17.8
Camera direction	Gemma-4-12B	25.2	22.3	—	5.9	+2.9
	Qwen2.5-VL-7B	24.7	29.2	—	25.4	-4.5
	Qwen3.5-4B	21.4	20.7	—	25.2	+0.7
	Qwen3.5-9B	23.5	25.7	—	26.1	-2.2

full—best1 = full — max(sat, sv) · camera_direction has no sv_only (sv = —)

Which tasks need both views?

full – best single view · Qwen3.5-9B · by task family

family	task	full–best1	reading
cross-view	Mismatch binary (easy)	+33.0	needs both
cross-view	Mismatch binary (hard)	+17.8	needs both
cross-view	Mismatch MCQ (easy)	+27.8	needs both
cross-view	Mismatch MCQ (hard)	+17.8	needs both
cross-view	Camera direction	-2.2	near chance
urbanization	Land use	+0.3	marginal
urbanization	Building height	-3.6	2nd view redundant
urbanization	Urban density	-6.4	2nd view redundant
urbanization	Junction type	+0.0	2nd view redundant
urbanization	Green space	-1.4	2nd view redundant
urbanization	Amenity richness	+0.7	marginal
urbanization	Road type	-10.0	2nd view redundant
urbanization	Road surface	-4.9	2nd view redundant
urbanization	Transit density	+1.2	marginal

cross-view from 4-model sweep (Qwen3.5-9B) · urbanization from strategy run (Qwen3.5-9B, thinking off)